

The Kirchberg–Rørdam character criterion

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Abstract

We prove the Kirchberg–Rørdam character criterion: for every free ultrafilter on $\mathbb{N}$, a nonzero unital separable complex C^* -algebra absorbs the Jiang–Su algebra if and only if its norm central-sequence algebra has no characters. We also prove that the infinite minimal tensor power of every nonzero unital separable complex C^* -algebra without characters is $\mathcal{Z}$ -stable and contains a unital copy of $\mathcal{Z}$.

1 Introduction

For a nonzero unital C^* -algebra A and a free ultrafilter ω on $\mathbb{N}$, write

$$A_\omega = \ell^\infty(\mathbb{N}, A) / \{(a_n) : \lim_{n \rightarrow \omega} \|a_n\| = 0\}, \quad F_\omega(A) = A_\omega \cap A',$$

where A is identified with its diagonal image in A_ω . A *character* is a nonzero $*$ -homomorphism to $\mathbb{C}$; on a unital algebra it is automatically unital. We write $\mathcal{Z}$ for the Jiang–Su algebra introduced by Jiang and Su [JS]. An algebra is *$\mathcal{Z}$ -stable* if it is isomorphic to its minimal tensor product with $\mathcal{Z}$.

Tensorial absorption can be recognized through central copies of the absorbed algebra. The theory of strongly self-absorbing algebras developed by Toms and Winter makes this principle precise [TW]. In the case of $\mathcal{Z}$, Kirchberg and Rørdam proved that a unital separable algebra is $\mathcal{Z}$ -stable exactly when its norm central-sequence algebra contains, with the same unit, a separable subhomogeneous algebra without characters [KR, Theorem 2.5]. Here subhomogeneity means a common finite bound on the dimensions of irreducible representations. The absence of characters on that subalgebra plainly forces the same absence on the whole central-sequence algebra. The question is whether this elementary obstruction is already sufficient.

Kirchberg and Rørdam asked whether the absence of characters on $F_\omega(A)$ characterizes Jiang–Su absorption among unital separable C^* -algebras [KR, Question 3.1]. Their work relates this question to tensor powers of algebras without characters [KR, Question 3.7 and Proposition 3.9]. The character question also appears as Problem LXXX in the recent collection of Schafhauser, Tikuisis, and White [STW, Section 22]. We prove the proposed characterization.

Theorem 1.1. *Let A be a nonzero unital separable complex C^* -algebra, and let ω be any free ultrafilter on $\mathbb{N}$. Then*

$$F_\omega(A) \text{ has no characters} \iff A \cong A \otimes_{\min} \mathcal{Z}.$$

The hypothesis uses the norm central-sequence algebra. The theorem applies to all such A , with no nuclearity, simplicity, trace, dimension, or comparison assumption. Its main ingredient is the following finite construction, in which separability is unnecessary.

Theorem 1.2. *Let D be a nonzero unital complex C^* -algebra without characters. There is an integer $m \geq 1$ and a unital $*$ -homomorphism*

$$I(2, 3) \longrightarrow D^{\otimes_{\max} m},$$

where

$$I(2, 3) = \{f \in C([0, 1], M_2 \otimes M_3) : f(0) \in M_2 \otimes 1_3, f(1) \in 1_2 \otimes M_3\}.$$

The exponent depends on D . The conclusion is a unital homomorphism, not a claim that this finite tensor power itself absorbs $\mathcal{Z}$. Its image is subhomogeneous and characterless, even if the homomorphism has a kernel. These are precisely the properties needed for the central-sequence application. They also give the following consequence.

Corollary 1.3 (Infinite minimal tensor powers). *Let D be a nonzero unital separable complex C^* -algebra without characters, and write*

$$B = D^{\otimes_{\min} \infty} := \varinjlim_{n \geq 1} (D^{\otimes_{\min} n}, x \longmapsto x \otimes 1_D).$$

Then $B \cong B \otimes_{\min} \mathcal{Z}$, and B contains a unital copy of $\mathcal{Z}$.

Separability is needed for this application of Dadarlat–Toms, but neither nuclearity nor simplicity is assumed. The absorption conclusion concerns the infinite minimal tensor power, not a finite power.

1.1 Earlier results and the remaining construction

Dadarlat and Toms connected Jiang–Su absorption with infinite tensor powers [DT]. Their Theorem 1.1 shows that the infinite minimal tensor power of a unital separable algebra is $\mathcal{Z}$ -stable whenever it contains a same-unit subhomogeneous characterless subalgebra. The construction in their Theorem 6.2 gives, for each fixed prime dimension-drop algebra, a unital map into any separable recursive subhomogeneous algebra of finite topological dimension whose minimum matrix size is sufficiently large relative to that dimension. Our finite theorem begins only with absence of characters on D and uses maximal tensor products, so it must construct the required subhomogeneous image without such a structural hypothesis.

A positive element is *full* if the closed two-sided ideal it generates is the whole ambient algebra. Several regularity criteria explain why finding the subhomogeneous image is useful. Robert and Tikuisis showed that two full orthogonal positive elements in the central-sequence algebra suffice for absorption when the original separable algebra has finite nuclear dimension [RT, Theorem 1.2]. More recently, Perera, Thiel, and Vilalta characterized $\mathcal{Z}$ -stability of separable algebras by purity of the central-sequence algebra [PTV, Theorem 3.6]. Purity imposes comparison and divisibility conditions on its Cuntz semigroup. The present argument assumes neither finite nuclear dimension nor these comparison and divisibility properties.

The companion [OAI, Theorem 1.4] proves that a finite maximal tensor power of every nonzero unital characterless algebra contains two full orthogonal positive elements, without separability or nuclearity. Its Gaussian and pure-state-excision proof is different from the representation-label argument developed below. We also use the companion’s commuting-copy lemma [OAI, Lemma 9.3], with the primary reindexing provenance in [KR, Proposition 2.2 and Corollary 2.3].

The bridge from full orthogonal elements to a homomorphism from the matrix cone $C_0((0, 1]) \otimes M_2$ that sends $t \mapsto te_{11}$ to a full element already appears in the proof of [KR, Proposition 6.1].

We give a complete construction using representation-stable tensor labels. A matrix-cone map can equivalently be viewed as a completely positive contractive order-zero map on a matrix algebra, meaning one that preserves orthogonality of positive elements, as in Winter and Zacharias [WZ, Corollary 4.1]; below we construct the cone homomorphism directly. Fullness of its first matrix corner is not by itself the dimension-drop criterion. The additional task is to produce two *exactly equivalent* orthogonal positive contractions, of the form v^*v and vv^* for a single element v , and compare their unit defect with a positive cutoff of one of them. This is the input to the stable-rank-free implication of Rørdam and Winter [RW, Proposition 5.1, (ii) $\Rightarrow$ (iv)], whose needed form is proved explicitly in Section 3.

1.2 Proof overview

We give the finite construction in two stages. First, a finite-dimensional space of selfadjoints detects noncommutativity in every representation of D . An incidence argument supplies tensor labels that remain linearly independent after arbitrary rank-at-least-two maps on the individual legs. These labels yield two selfadjoints that do not commute in any irreducible representation of a suitable tensor power. A double commutator then has full positive and negative parts, and a second use of the labels assembles finitely many bridges between them into one full square-zero element.

Second, the square-zero element gives a matrix-cone homomorphism with a full first corner. A continuous simplex of embeddings of M_2 organizes this cone across a finite tensor power. Two orthogonal equivalent positive elements leave a defect concentrated near the origin of a cube. Disjoint regions near its coordinate axes support elements that absorb this defect after a further finite amplification. The resulting comparison is precisely the one required by the Rørdam–Winter dimension-drop criterion. This comparison is the step that turns full orthogonal elements into a unital $I(2, 3)$ map.

Section 2 records the comparison conventions and structural criteria, with the dimension-drop conversion proved in Section 3. Section 4 constructs this cone homomorphism, and Section 5 proves the finite comparison. Section 6 deduces both theorems and Corollary 1.3. All tensor exponents and comparison constants are allowed to depend on the algebra D .

2 Preliminaries

2.1 Tensor products and positive elements

All algebras are complex. All tensor products in the construction are *maximal*, and we abbreviate $D^{\otimes_{\max} n}$ to $D^{\otimes n}$. We specify minimal tensor products explicitly. Algebraic tensor products are denoted by $\odot$ over $\mathbb{C}$ and by $\otimes_{\mathbb{R}}$ over $\mathbb{R}$. Representations called irreducible are nonzero and nondegenerate, hence unital when the represented algebra is unital. An element is *full* if its generated closed two-sided ideal is the whole algebra. Equivalently, it is nonzero in every irreducible representation: a proper ideal has a nonzero irreducible representation of its quotient.

For positive elements in finite matrix algebras over a unital algebra B , write $a \precsim b$ if, after adjoining zero corners as necessary, there are matrices x_j with $x_j^* b x_j \rightarrow a$ in norm. Write $[a]$ for the resulting Cuntz equivalence class. Thus $[a] \leq [b]$ means $a \precsim b$, and addition is induced by direct sum. We use only the following elementary rules:

1. $[v^*v] = [vv^*]$ for every rectangular matrix v ;
2. if $a_1, \dots, a_l$ are positive and pairwise orthogonal, then $[\sum_i a_i] = \sum_i [a_i]$;

3. comparison and addition are preserved by $*$ -homomorphisms and by external tensoring with a fixed positive element, including for maximal tensor products.

For the first rule, with $b = v^*v$, the matrices $x_\gamma = v(b + \gamma)^{-1/2}$ satisfy $x_\gamma^*(vv^*)x_\gamma = b^2(b + \gamma)^{-1} \rightarrow b$ as $\gamma \downarrow 0$; apply the same argument to v^* for the converse. For the second rule, apply the first to the row $(a_1^{1/2}, \dots, a_l^{1/2})$. For the third rule, apply a homomorphism to the witnesses, or tensor them with the unit of the other algebra. The norm estimate for the resulting error is at most the original error times the norm of the fixed positive tensor factor. Direct sums agree under the canonical matrix identifications. These rules use no cancellation or regularity of the Cuntz order. We also use $[a] \leq [1_B]$ for $a \in B_+$, which follows from the exact witness $a^{1/2}$.

2.2 The dimension-drop criterion

We say that positive elements $a_1, a_2 \in B$ are *equivalent* if $a_1 = v^*v$ and $a_2 = vv^*$ for one $v \in B$. This is the exact relation required in the next proposition.

Proposition 2.1 (Rørdam–Winter). *Let B be a unital C^* -algebra. Suppose that $a_1, a_2 \in B$ are orthogonal positive contractions with $a_1 = v^*v$, $a_2 = vv^*$ for some $v \in B$, and that, for some $\varepsilon > 0$,*

$$1_B - a_1 - a_2 \precsim (a_1 - \varepsilon)_+.$$

Then there is a unital $$ -homomorphism $I(2, 3) \rightarrow B$.*

This is [RW, Proposition 5.1, (ii) $\Rightarrow$ (iv), with $n = 2$]. Section 3 supplies a complete proof using their method, with the support projections, scalar endpoint fiber, and weighted flip identity made explicit. The stable-rank-one hypothesis appearing elsewhere in that proposition is needed only for (i) $\Rightarrow$ (ii). It is not a hypothesis of the implication stated here. Orthogonality ensures $a_1 + a_2 \leq 1_B$, so the defect in the display is positive.

2.3 Central-sequence inputs

We use the following results of Kirchberg and Rørdam for the fixed free ultrafilter ω .

Proposition 2.2 (Kirchberg–Rørdam). *Let A be a nonzero unital separable C^* -algebra.*

1. *If $D \subset F_\omega(A)$ is separable and contains the ambient unit, then for every $m \geq 1$ there is a unital $*$ -homomorphism $D^{\otimes_{\max} m} \rightarrow F_\omega(A)$.*
2. *$A \cong A \otimes_{\min} \mathcal{Z}$ if and only if $F_\omega(A)$ contains, with the ambient unit, a separable subhomogeneous C^* -algebra without characters.*

The first assertion is the unital specialization of the commuting-copies lemma in [OAI, Lemma 9.3], whose primary source is [KR, Proposition 2.2 and Corollary 2.3]. For unital A , the annihilator ideal used in that companion is zero: an element annihilating every element of A annihilates its unit. Thus its $F(A)$ is precisely the norm central-sequence algebra used here. This particular lemma has no nuclear-dimension assumption. We include the short specialization to make its quantifiers explicit.

Proof of Proposition 2.2(1). For any separable subalgebra $E \subset F_\omega(A)$, we construct a unital map from D into $F_\omega(A) \cap E'$. Choose a countable dense unital $*$ -subalgebra $D_0 \subset D$ over $\mathbb{Q} + i\mathbb{Q}$. For every $x \in D_0$, choose a bounded lift $(a_l^x)_l$ in A with $\sup_l \|a_l^x\| \leq 2\|x\|$, using the zero lift for $x = 0$

and the constant unit lift for $x = 1$. Such choices follow from the quotient norm, by truncating coordinate norms without changing the ultrapower class. All additive, rational-complex scalar, multiplicative and adjoint relation defects tend to zero in norm along ω .

Choose a countable dense subset of A and bounded lifts $(b_s^j)_s$ of a countable dense subset of E . Enumerate the relation-defect tests and the tests for centrality of the a_l^x against the chosen subset of A . For each output index s , choose $l(s)$ so that the first s tests hold within $1/s$, and so that

$$\|[a_{l(s)}^x, b_s^j]\| < 1/s$$

for the first s pairs (x, j) in an enumeration. Each of the finitely many requirements holds on an ω -large set of old indices l : for this fixed s , each b_s^j is a fixed element of A , so original centrality applies. Their finite intersection is nonempty.

The new sequences $(a_{l(s)}^x)_s$ are central and satisfy every algebraic relation in the ordinary limit, hence along the arbitrary free ultrafilter ω . Their classes define a unital rational-complex $*$ -homomorphism on D_0 into $F_\omega(A) \cap E'$. The uniform bound $2\|x\|$ extends it continuously to a complex unital $*$ -homomorphism on D . Starting with the given copy of D and taking E successively to be the algebra generated by the earlier images produces any prescribed finite number of commuting unital images. The universal property of the maximal tensor product gives (1). Injectivity of the reindexed maps is not needed. $\square$

The second assertion is [KR, Theorem 2.5]. Here *subhomogeneous* means that the dimensions of all irreducible representations have a common finite bound. In particular, this input does not replace norm central sequences by a tracial ultrapower or impose nuclearity on A .

3 The dimension-drop criterion without stable rank one

This section proves the implication of Rørdam and Winter [RW, Proposition 5.1, (ii) $\Rightarrow$ (iv)] used in the finite construction. We retain their order-zero and flip-path method, while making the support projections and endpoint domains explicit. The argument applies to an arbitrary unital C^* -algebra.

For $b \geq 0$, write $\text{Her}(b) = \overline{bBb}$. Support projections and polar decompositions will be taken in B^{**} . They need not belong to B ; we will verify membership in B whenever they are used to construct an element of the original algebra.

Lemma 3.1. *If $a, b \in B_+$ satisfy $a \precsim b$, then for every $\delta > 0$ there is $z \in B$ such that*

$$z^*z = (a - \delta)_+, \quad zz^* \in \text{Her}(b).$$

Proof. Since a, b lie in B , compression of stabilized comparison witnesses to their $(1, 1)$ entries gives witnesses in B itself. Choose $x \in B$ with $\|a - x^*bx\| < \eta < \delta$, and put $e = (a - \delta)_+$ and $y = b^{1/2}xe^{1/2}$. Then

$$T = y^*y \geq e^{1/2}(a - \eta)e^{1/2} \geq (\delta - \eta)e, \quad T \leq \|x^*bx\|e.$$

Thus T and e have the same support. If $y = uT^{1/2}$ is the polar decomposition, set $z = ue^{1/2}$ in B^{**} . To see that $z \in B$, approximate it by

$$z_\tau = y(T + \tau)^{-1/2}e^{1/2} \in b^{1/2}B, \quad \tau > 0.$$

Writing $f_\tau(t) = 1 - \sqrt{t/(t+\tau)}$, the domination of e by T gives

$$\|z - z_\tau\|^2 \leq \|f_\tau(T)ef_\tau(T)\| \leq \frac{\sup_{t \geq 0} tf_\tau(t)^2}{\delta - \eta} \leq \frac{\tau}{\delta - \eta}.$$

No commutation of e with T is used here. Hence $z \in B$, $z^*z = e$, and zz^* is a norm limit of elements of $b^{1/2}Bb^{1/2}$, as required. $\square$

We shall use a concrete description of order-zero maps. If $x_1, \dots, x_q$ are contractions with $x_i x_j^* = \delta_{ij} h$ for a positive contraction h , then

$$\alpha(e_{ij}) = x_i^* x_j$$

defines a completely positive contractive order-zero map from M_q . Indeed, complete positivity follows from this Gram formula. The positive elements $x_i^* x_i$ are orthogonal, so their sum H is a contraction; multiplication gives $\alpha(e_{ij})\alpha(e_{kl}) = \delta_{jk} H \alpha(e_{il})$. This identity proves preservation of orthogonality. It also gives a supporting homomorphism $\bar{\alpha} : M_q \rightarrow B^{**}$ with

$$\alpha(x) = H\bar{\alpha}(x) = \bar{\alpha}(x)H, \quad \bar{\alpha}(1_q) = \text{supp}(H).$$

For completeness, its matrix units are the strong limits of $\alpha(e_{ij})(H + \tau)^{-1}$ as $\tau \downarrow 0$. The displayed multiplication identity and $\alpha(e_{ij})^* \alpha(e_{ij}) \leq H^2$ verify their boundedness, matrix-unit relations, and stated support.

Lemma 3.2. *Let $\alpha : M_3 \rightarrow B$ and $\beta : M_2 \rightarrow B$ be completely positive contractive maps with commuting ranges, satisfying $\psi(x)\psi(y) = \psi(1)\psi(xy)$ for $\psi = \alpha, \beta$. If $\alpha(1_3) + \beta(1_2) = 1_B$, there is a unital homomorphism $I(2, 3) \rightarrow B$.*

Proof. This is the order-zero presentation used in [RW, Proposition 2.5]; we include the verification. The canonical maps into $I(2, 3)$ are

$$\alpha_0(y)(t) = t(1_2 \otimes y), \quad \beta_0(x)(t) = (1 - t)(x \otimes 1_3).$$

Their generated algebra contains the scalar function t , and products of their entries generate the ideal $C_0((0, 1), M_2 \otimes M_3)$. Their endpoint values also generate $M_2 \oplus M_3$. Consequently they generate all of $I(2, 3)$.

Put $E = C^*(\alpha(M_3), \beta(M_2), 1_B)$. The element $\alpha(1_3)$ is central in E . In any irreducible representation of E it is a scalar $r \in [0, 1]$. If $0 < r < 1$, the rescaled maps $r^{-1}\alpha$ and $(1 - r)^{-1}\beta$ in this representation are commuting unital matrix homomorphisms, and hence define a representation of $M_2 \otimes M_3$. At $r = 0$ only the M_2 map remains, and at $r = 1$ only the M_3 map remains. These assertions follow either from the supporting homomorphisms or directly from the order-zero multiplication identity.

It follows that substitution of α, β for α_0, β_0 in a noncommutative $*$ -polynomial is bounded in norm by the norm of the polynomial in $I(2, 3)$: test in each irreducible representation of E and use the corresponding evaluation at r . Substitution is therefore well defined and extends from the dense generated algebra to the required unital homomorphism. $\square$

Proof of Proposition 2.1. Exact relations from the comparison. Write $a_1 = w^*w$ and $a_2 = ww^*$ for the equivalence witness. Decreasing ε if necessary, assume $0 < \varepsilon < 1$. Let $f(t) = \min\{t/\varepsilon, 1\}$ and put $s_1 = f(a_1)^{1/2}$. If $w^* = u_0 a_2^{1/2}$ is its polar decomposition, put $s_2 = u_0 f(a_2)^{1/2}$. This belongs to

B : it equals $w^*q(a_2)$, where the continuous function $q(t) = \sqrt{f(t)}/t$ for $t > 0$ is assigned the value $\varepsilon^{-1/2}$ at zero. Thus

$$s_i s_j^* = \delta_{ij} s_1^2, \quad d := s_1^* s_1 + s_2^* s_2 = f(a_1) + f(a_2).$$

Here the off-diagonal products vanish because the initial positive elements are orthogonal. Writing $c = 1 - a_1 - a_2$, functional calculus in the commuting algebra generated by a_1, a_2 gives

$$a := 1 - d = \varepsilon^{-1}(c - (1 - \varepsilon))_+.$$

Apply Lemma 3.1 to $c \preceq (a_1 - \varepsilon)_+$, with $\delta = 1 - \varepsilon$, and divide its element by $\sqrt{\varepsilon}$. We obtain a contraction $v \in B$ satisfying

$$v^* v = a = 1 - d, \quad vv^* \in \text{Her}((a_1 - \varepsilon)_+), \quad s_1^2 vv^* = vv^*. \quad (3.1)$$

In particular $s_1 v = v$, $s_2 v = 0$, and $av = 0$. The initial and final support relations also give $ds_i = s_1^2 s_i = s_i d$; hence a commutes with the s_i . The Gram construction gives $\rho : M_2 \rightarrow B$ with $\rho(e_{ij}) = s_i^* s_j$ and $\rho(1_2) = d$. Write σ for its supporting homomorphism and $p = \text{supp}(d)$.

Two orthogonal copies of the defect algebra. Let $v = ua^{1/2}$ be the polar decomposition and $e = u^* u = \text{supp}(a)$. Set

$$r_i = s_i^* u = \sigma(e_{i1})u \quad (i = 1, 2), \quad R = r_1 r_1^* + r_2 r_2^*.$$

The second formula for r_i follows from $du = u$ and $s_1 u = u$. Direct multiplication gives

$$r_i^* r_j = \delta_{ij} e, \quad dr_i = r_i, \quad R \leq p, \quad R \perp e, \quad \sigma(e_{jk})r_i = \delta_{ki} r_j. \quad (3.2)$$

For $R \perp e$, note that a commutes with each s_i (because d does) and $au = 0$. The map

$$\varphi : \text{Her}(a) \longrightarrow B, \quad \varphi(b) = \sum_{i=1}^2 r_i b r_i^*$$

is a homomorphism by (3.2). Its values belong to B : approximate $b \in \text{Her}(a)$ by $a^{1/2} b_m a^{1/2}$ and use $ua^{1/2} = v$ to see that $ub u^* \in B$. The same approximation, with one outer factor omitted, shows that $ub \in B$. The same relations show

$$\varphi(b)r_i = r_i b, \quad [\varphi(b), \sigma(M_2)] = 0, \quad \varphi(\text{Her}(a)) \perp \text{Her}(a). \quad (3.3)$$

Here $\varphi(b) = \sum_i r_i b r_i^*$ places the same defect element in the two ranges of the r_i , while $\sigma(e_{jk})$ acts on their index i . To obtain the other coordinate, we use sections on $[0, 1]$ with values in M_2 , evaluated at d , that are scalar at zero and vanish at one. The scalar value accounts for the corner outside $p = \text{supp}(d)$; vanishing at one keeps the evaluated section in $\text{Her}(1 - d)$. Define

$$D_1 = \{F \in C([0, 1], M_2) : F(0) \in \mathbb{C}1_2, F(1) = 0\},$$

and let c_F be the scalar with $F(0) = c_F 1_2$. Since d commutes with $\sigma(M_2)$, continuous functional calculus defines $\bar{\lambda}(F) = \sum_{i,j} F_{ij}(d) \sigma(e_{ij})$ in $pB^{**}p$. The required homomorphism is

$$\lambda(F) = \bar{\lambda}(F) + c_F(1 - p). \quad (3.4)$$

The extra summand retains the scalar fiber at zero. In particular $\lambda(f1_2) = f(d)$ for scalar f vanishing at one. The algebra D_1 is generated by $(1 - t)1_2$ and $t(1 - t)x$, $x \in M_2$. Their images

are respectively a and $a\rho(x)$, so λ takes values in $\text{Her}(a)$. This also verifies membership in B , without assuming $p \in B$.

Let

$$D_2 = \{F \in C([0, 1], M_2 \otimes M_2) : F(0) \in \mathbb{C}1_4, F(1) = 0\}.$$

It is a subalgebra of $D_1 \otimes M_2$. By (3.3), the commuting product of $\varphi \circ \lambda$ and σ defines a homomorphism on $D_1 \otimes M_2$; denote its restriction to D_2 by γ . Its range lies in B : since $d\varphi(b) = \varphi(b)$, $\varphi(b)\sigma(x) = \varphi(b)\rho(x) \in B$. Its support projection is R , because its range is supported there and contains $\varphi(a)$, which has support R .

The product defining γ pairs this transported section action with the action of σ on the copy index. A weighted flip will interchange them only on the overlap needed for the final commutation calculation.

The flip where the two pieces overlap. Let $F \in M_2 \otimes M_2$ be the flip unitary. Choose a continuous function $\theta : [0, 1] \rightarrow [0, 1]$ that is zero on $[0, 1/3]$ and one on $[2/3, 1]$, and set

$$U(t) = \frac{1+F}{2} + e^{i\pi\theta(t)} \frac{1-F}{2}.$$

This is an ordinary unitary path, equal to 1 near zero and to F on $[2/3, 1]$. It is a multiplier of D_2 . The multiplier extension of γ sends it to a unitary w_0 in the corner $RB^{**}R$; put $W = w_0 + 1 - R$.

Let $g(t) = 1$ for $0 \leq t \leq 2/3$ and $g(t) = 3(1-t)$ otherwise. Put

$$h = \lambda(g1_2) = g(d), \quad k = \varphi(h), \quad l = 1 - h - k.$$

The elements h, k are orthogonal positive contractions, and $g(t) \geq 1-t$ implies $0 \leq l \leq d$. Also $kW = Wk$ and

$$k^\alpha W = \gamma(g^\alpha U) \in B \quad (\alpha > 0).$$

The formula is legitimate because $g^\alpha U \in D_2$. Define

$$x_i = h^{1/2} r_i^* W \quad (i = 1, 2), \quad x_3 = h^{1/2}.$$

These elements belong to B : for $i \leq 2$ use the factorization

$$x_i = (r_i^* k^{1/4})(k^{1/4} W), \quad r_i^* k^{1/4} = h^{1/4} r_i^* \in B.$$

By (3.2), $R \perp e$, and unitarity of W ,

$$x_i x_j^* = \delta_{ij} h \quad (1 \leq i, j \leq 3), \quad \sum_{i=1}^3 x_i^* x_i = h + k.$$

Thus $\alpha(e_{ij}) = x_i^* x_j$ defines an order-zero map $\alpha : M_3 \rightarrow B$ with $\alpha(1_3) = h + k$.

Since h, k commute with $\sigma(M_2)$ and $0 \leq l \leq d$, the formula $\beta(y) = l\sigma(y)$ defines an order-zero map $\beta : M_2 \rightarrow B$. To justify its range directly, l vanishes on the spectral subspace of d for $[0, 2/3]$, since $h = 1$ there and $hk = 0$. Choose a continuous scalar function j that equals $1/t$ for $t \geq 2/3$. Then $\beta(y) = lj(d)\rho(y) \in B$. Since $lp = l$, we have $\alpha(1_3) + \beta(1_2) = 1$.

It remains to check that their ranges commute. This is precisely where the flip is needed. The scalar function $Q = g(1-g)$ vanishes at both endpoints and is supported where $U = F$. Therefore

$$Q(t)(1 \otimes y)U(t)^* = U(t)^*Q(t)(y \otimes 1), \quad y \in M_2.$$

Both sides are elements of D_2 . Applying γ gives the well-defined identity

$$k(1-k)\sigma(y)W^* = W^*\varphi(\lambda(Q \otimes y)). \quad (3.5)$$

In particular, no constant nonscalar section is evaluated by λ . For $i = 1, 2$, put $C_i = \alpha(e_{i3}) = W^*r_ih$. Using (3.3) and (3.5), we obtain

$$\begin{aligned} \beta(y)C_i &= k(1-k)\sigma(y)W^*r_i \\ &= W^*r_i\lambda(Q \otimes y) \\ &= W^*r_ih(1-h)\sigma(y) = C_i\beta(y). \end{aligned}$$

Here $\lambda(Q \otimes y) = Q(d)\sigma(y)$ because $Q(0) = 0$; the last equality uses $hk = 0$.

The C^* -algebra generated by the C_i contains $\alpha(M_3)$. Indeed, $C_i^*C_i = h^2$ and $C_iC_i^* = \alpha(e_{ii})^2$, so their generated algebra contains h , all diagonal entries, and $A_0 := \alpha(1_3)$. Moreover $C_iC_j^* = A_0\alpha(e_{ij})$, and multiplication by $(A_0 + \tau)^{-1}$ followed by $\tau \downarrow 0$ recovers every remaining matrix entry. The convergence follows from $\alpha(e_{ij}) = A_0\bar{\alpha}(e_{ij})$ and $\|\tau A_0(A_0 + \tau)^{-1}\| \leq \tau$. Thus $\alpha(M_3)$ commutes with $\beta(M_2)$. Lemma 3.2 completes the proof.

The construction also covers $v = 0$: then $a = h = k = 0$, $R = 0$, $W = 1$, and β is a unital M_2 homomorphism. No step requires stable rank one or a nonzero defect. $\square$

4 A matrix-cone homomorphism with a full corner

We first construct a full square-zero element in a finite tensor power of a unital C^* -algebra without characters. Its polar decomposition will then give the matrix-cone homomorphism needed in the next section. The absence of characters is detected by finitely many commutators. The main issue is to combine those finitely many tests into one element that survives every irreducible representation of a maximal tensor product, including representations that are not spatial tensor products.

4.1 Algebraic independence of commuting factors

Lemma 4.1. *Let $M \subseteq B(H)$ be a von Neumann factor with $1_M = 1_H$, and let $B \subseteq M'$ be a unital algebra. The multiplication map*

$$M \odot B \longrightarrow B(H), \quad a \otimes b \longmapsto ab,$$

is injective. Consequently, multiplication is injective on the complex algebraic tensor product of any finite family of commuting von Neumann factors on the same Hilbert space, all with unit 1_H .

Proof. Fix complex linearly independent elements $b_1, \dots, b_l \in B$. It suffices to show that

$$S = \left\{ (a_1, \dots, a_l) \in M^{1 \times l} : \sum_{i=1}^l a_i b_i = 0 \right\}$$

is zero. This row space is weak-operator closed and invariant under diagonal multiplication by M on either side. The right invariance uses $B \subseteq M'$. Let $J \subseteq M_l(M)$ consist of the matrices whose rows belong to S . It is a weak-operator closed left ideal: each row of a left matrix product is an M -linear combination of the original rows.

We give the support-projection argument for the form of this ideal. For $x \in J$, we have $x^*x \in J$. Functional calculus with functions vanishing at zero, followed by weak-operator closure, puts the support projection of x^*x in J : one may first take $x^*x(x^*x + \varepsilon 1)^{-1}$ and then let $\varepsilon \downarrow 0$. Finite joins of these projections also belong to J , since the join of finitely many projections is the support of their sum. Their increasing join p therefore belongs to J ; this is a net of finite joins, so no separability assumption is involved. Every $x \in J$ satisfies $x = xp$, and $p \in J$ gives

$$J = M_l(M)p.$$

For any unitary $u \in M$, put $U = \text{diag}(u, \dots, u)$. Right invariance and its counterpart for u^* give $JU = J$. The right support projection of xU is the conjugate by U of the right support projection of x . Taking their joins gives

$$U^*pU = p.$$

Thus every entry of p belongs to the center of M , and is scalar. Each row of p is consequently a scalar relation among the independent b_i , so $p = 0$. Hence $J = 0$ and $S = 0$, proving the first assertion.

For finitely many commuting factors, apply the first assertion to one factor and the algebra generated by the remaining factors, and induct on their number. $\square$

Lemma 4.2. *Let $\pi : B_1 \otimes_{\max} \cdots \otimes_{\max} B_q \rightarrow B(H)$ be an irreducible representation, where the B_j are nonzero unital C^* -algebras. The weak-operator closures of the represented tensor legs are commuting von Neumann factors. The same is true after grouping the legs according to any partition into nonempty blocks.*

Proof. Every represented block contains $\pi(1) = 1_H$. The represented blocks commute, and this commutation persists under weak-operator closure. A central element of the closure of one block commutes both with that block and with every complementary block. It therefore belongs to the commutant of the full irreducible image, which consists of the scalars. This proves factorhood for each block. $\square$

These lemmas concern algebraic tensor independence. They require neither faithfulness of the restrictions to the tensor legs nor a spatial decomposition of the represented factors.

4.2 Labels that remain independent in representations

Lemma 4.3. *Let D be a nonzero unital C^* -algebra without characters. There is a finite-dimensional real subspace $V \subseteq D_{\text{sa}}$ such that the image of V contains a noncommuting pair in every nonzero unital representation of D .*

Proof. The closed ideal generated by the commutators of selfadjoint elements is all of D . Otherwise the quotient by this ideal would be a nonzero unital commutative C^* -algebra, and a character of the quotient would give a character of D .

Choose a finite algebraic expression in these commutators within distance less than one of 1_D . This expression is invertible, so the finitely many commutators it uses already generate the unit ideal. Let V be the real span of their selfadjoint entries. If a unital representation sent V to a commuting space, it would annihilate all these commutators, and hence their unit ideal. This is impossible. $\square$

Lemma 4.4. *Let V be a finite-dimensional real vector space with $m = \dim_{\mathbb{R}} V \geq 2$. For every integer $r \geq 1$, there are $n \geq 1$ and tensors $x_1, \dots, x_r \in V^{\otimes_{\mathbb{R}} n}$ with the following property: whenever $P_j : V \rightarrow E_j$ are real linear maps of rank at least two, the tensors*

$$(P_1 \otimes \cdots \otimes P_n)(x_i), \quad 1 \leq i \leq r,$$

are linearly independent over $\mathbb{R}$.

Proof. It suffices to consider surjections $P_j : V \rightarrow \mathbb{R}^2$. Indeed, a map of rank at least two can be postcomposed, on its finite-dimensional image, with a surjection onto $\mathbb{R}^2$; a dependence before this postcomposition would remain a dependence afterward.

Choose n such that

$$2^n > 2mn + r - 1,$$

and put $W = V^{\otimes_{\mathbb{R}} n}$. Let $\mathcal{P} = \text{Surj}(V, \mathbb{R}^2)^n$, an open subset of a real vector space of dimension $2mn$. Consider the smooth map

$$\begin{aligned} \Phi : \mathcal{P} \times S^{r-1} \times W^r &\longrightarrow (\mathbb{R}^2)^{\otimes_{\mathbb{R}} n}, \\ (P, u, x) &\longmapsto (P_1 \otimes \cdots \otimes P_n) \left(\sum_{i=1}^r u_i x_i \right). \end{aligned}$$

Its differential in the x variables is surjective everywhere: some $u_i \neq 0$, and the tensor product of the P_j is surjective. Thus $\Phi^{-1}(0)$ is a smooth manifold of dimension

$$2mn + r - 1 + rm^n - 2^n < rm^n = \dim_{\mathbb{R}} W^r.$$

By Sard's theorem, its projection to W^r has Lebesgue measure zero. The manifolds here are second countable, so this assertion can be applied on a countable atlas; no compactness of $\mathcal{P}$ is needed. Choose $x \in W^r$ outside that projection. For this one tuple, no nonzero real dependence occurs for any tuple of surjections: any nonzero coefficient vector can be normalized to lie in S^{r-1} . $\square$

Corollary 4.5. *Let D and V be as in Lemma 4.3. For each $r \geq 1$, there are $n \geq 1$ and selfadjoint elements $x_1, \dots, x_r \in D^{\otimes n}$ whose images are complex linearly independent under every irreducible representation of $D^{\otimes n} \otimes_{\max} B$, for every nonzero unital C^* -algebra B . Here the elements x_i occupy the first n tensor legs.*

Proof. Take the real tensors supplied by Lemma 4.4 and map them naturally into $D^{\otimes n}$. They are selfadjoint, since they are real linear combinations of elementary tensors with selfadjoint entries.

Fix such an irreducible representation and write π_j for its restriction to the j -th copy of D . Each π_j is unital, so Lemma 4.3 implies $\dim_{\mathbb{R}} \pi_j(V) \geq 2$. Apply Lemma 4.4 to the real maps $V \rightarrow \pi_j(V)$.

The resulting independence also holds after complexification. Indeed, a real basis of a space of selfadjoint operators is complex linearly independent: taking adjoints of a complex relation separates its real and imaginary parts. Hence the complexification of each $\pi_j(V)$ embeds in the represented algebra. Tensoring these injections shows that the real independent tensors remain complex independent in the complex algebraic tensor product of the represented legs: the latter tensor product contains the complexification of the real tensor product injectively. By Lemma 4.2 their weak closures are commuting factors, and Lemma 4.1 preserves this independence under multiplication. $\square$

4.3 A full square-zero element

The existence of two full orthogonal positive elements in a finite maximal tensor power is already established in [OAI, Theorem 1.4]. The bridge construction below is related to Kirchberg–Rørdam’s proof of [KR, Proposition 6.1, (i) $\Rightarrow$ (v)], which combines finitely many bridges with orthogonal full tensor labels. Here we use representation-stable linear independence and give the full construction of the square-zero element needed for the cone map.

The labels allow linear independence to replace orthogonality when combining finitely many algebraic witnesses. We first construct two full orthogonal positive elements, and then use a second label family to retain all bridges between them in a single square-zero element.

Lemma 4.6. *If D is a nonzero unital C^* -algebra without characters, then some finite maximal tensor power P of D contains a full element w satisfying $w^2 = 0$ and $\|w\| = 1$.*

Proof. Choose V as in Lemma 4.3, and let $v_1, \dots, v_m$ be a real basis of V . Take the labels $x_1, \dots, x_m \in D^{\otimes n}$ from Corollary 4.5. In

$$P_0 = D \otimes_{\max} D^{\otimes n} \otimes_{\max} D^{\otimes n}$$

define selfadjoint elements

$$s = \sum_{i=1}^m v_i \otimes x_i \otimes 1, \quad t = \sum_{i=1}^m v_i \otimes 1 \otimes x_i.$$

We first show that their images do not commute in any irreducible representation π of P_0 . Denote the restrictions to the three displayed blocks by π_0, π_L, π_R . Their weak closures are commuting factors by Lemma 4.2, and each of the two label families has independent images by Corollary 4.5. The expansion

$$\pi([s, t]) = \sum_{i,j=1}^m \pi_0([v_i, v_j]) \pi_L(x_i) \pi_R(x_j)$$

and Lemma 4.1 show that $\pi([s, t]) = 0$ would imply $\pi_0([v_i, v_j]) = 0$ for all i, j . This contradicts the defining property of V for the unital representation π_0 . Independence of the images of the v_i themselves is not required.

Set $z = [s, [s, t]]$, which is selfadjoint. We claim that both z_+ and z_- are full in P_0 . Fix an irreducible representation π , and suppose first that $\pi(z) \geq 0$. For every vector ξ , the real function

$$f_\xi(\lambda) = \left\langle e^{i\lambda\pi(s)} \pi(t) e^{-i\lambda\pi(s)} \xi, \xi \right\rangle, \quad \lambda \in \mathbb{R},$$

is bounded and twice continuously differentiable, with

$$f_\xi''(\lambda) = - \left\langle e^{i\lambda\pi(s)} \pi(z) e^{-i\lambda\pi(s)} \xi, \xi \right\rangle \leq 0.$$

A bounded concave function on the whole real line is constant. Consequently

$$f_\xi'(0) = \langle i\pi([s, t]) \xi, \xi \rangle = 0 \quad \text{for every } \xi.$$

The operator $i\pi([s, t])$ is selfadjoint, so it must be zero, a contradiction. The assumption $\pi(z) \leq 0$ leads to the same contradiction using bounded convex functions. Thus both the positive and negative parts of $\pi(z)$ are nonzero. Functional calculus commutes with π , so z_+ and z_- survive every irreducible representation and are full.

Let J be the closed ideal generated by $z_-P_0z_+$. The algebraic two-sided ideals

$$I_{\pm} = \text{span}\{az_{\pm}b : a, b \in P_0\}$$

are both norm dense in P_0 . Their products lie in J , since

$$(az_-b)(cz_+d) = a(z_-bcz_+)d.$$

Approximating 1_{P_0} from both I_- and I_+ shows that $1_{P_0} \in J$, and hence $J = P_0$. An algebraic expression within distance less than one of 1_{P_0} uses only finitely many elements of $z_-P_0z_+$ and their adjoints. We can therefore choose

$$c_i = z_-d_iz_+, \quad 1 \leq i \leq r,$$

whose closed ideal is P_0 . Notice that $c_ic_j = 0$ for all i, j , because $z_+z_- = 0$.

Choose new labels $x'_1, \dots, x'_r \in D^{\otimes n'}$ from Corollary 4.5, and put

$$P = P_0 \otimes_{\max} D^{\otimes n'}, \quad w_0 = \sum_{i=1}^r c_i \otimes x'_i.$$

The identities $c_ic_j = 0$ give $w_0^2 = 0$. To prove fullness, take any irreducible representation of P , with restrictions ρ_0 to P_0 and ρ_1 to the new label block. The images $\rho_1(x'_i)$ are independent by Corollary 4.5, allowing P_0 as the additional algebra. The two grouped weak closures are factors by Lemma 4.2. Thus Lemma 4.1 implies

$$\sum_i \rho_0(c_i) \rho_1(x'_i) = 0 \implies \rho_0(c_i) = 0 \quad \text{for all } i.$$

The latter conclusion is impossible, since the c_i generate the unit ideal and ρ_0 is unital. Hence w_0 is full. In particular $w_0 \neq 0$, and $w = w_0/\|w_0\|$ has the required properties. The exponents n, n' , and hence the exponent of P , depend on the finite detecting space and the finite bridge family; no bound uniform over D is asserted. $\square$

4.4 The cone homomorphism

Let

$$C = \{f \in C([0, 1], M_2) : f(0) \in \mathbb{C}1_{M_2}\}, \quad \iota(t) = t.$$

Thus C is the unitization of $C_0((0, 1], M_2)$.

Proposition 4.7. *For every nonzero unital C^* -algebra D without characters, there are an integer $m \geq 1$, the maximal tensor power $P = D^{\otimes m}$, and a unital $*$ -homomorphism $\eta : C \rightarrow P$ such that $\eta(\iota e_{11})$ is full in P .*

Proof. Let P and w be supplied by Lemma 4.6. Take the polar decomposition $w = u|w|$ in P^{**} , and put $p = u^*u$, $q = uu^*$. Since $w^2 = 0$, the closure of the range of w lies in its kernel; hence $q \leq 1 - p$, since p is the projection onto the orthogonal complement of that kernel in a faithful normal representation of P^{**} . The elements

$$E_{11} = p, \quad E_{22} = q, \quad E_{21} = u, \quad E_{12} = u^*$$

are therefore matrix units with unit $p + q$.

The positive element $h = |w| + |w^*|$ has norm one, support $p + q$, and commutes with these matrix units. Indeed, the two summands have orthogonal supports, and the polar decomposition gives

$$hE_{11} = |w|, \quad hE_{22} = |w^*|, \quad hE_{21} = w, \quad hE_{12} = w^*.$$

Commuting functional calculus now defines a $*$ -homomorphism

$$\begin{aligned} \varphi : C_0((0, 1], M_2) &\longrightarrow P^{**}, \\ f \otimes e_{ij} &\longmapsto f(h)E_{ij}, \quad f \in C_0((0, 1]). \end{aligned}$$

It takes its values in P . To see this directly, approximate f uniformly by polynomials vanishing at zero. Each corresponding term $h^k E_{ij}$, $k \geq 1$, belongs to P , because $hE_{ij} \in P$. Thus $f(h)E_{ij} \in P$. Unitizing φ gives a unital homomorphism $\eta : C \rightarrow P$, and

$$\eta(\iota e_{21}) = w, \quad \eta(\iota e_{11}) = |w|.$$

No assertion that u belongs to P , or that $p + q = 1_P$, is needed for this unitization.

Finally, w and $|w|$ generate the same closed ideal. The ideal of w contains w^*w and hence $|w|$ by functional calculus. Conversely,

$$w(|w| + \varepsilon 1_P)^{-1}|w| \longrightarrow w \quad (\varepsilon \downarrow 0)$$

in norm, so the ideal of $|w|$ contains w . Fullness of w therefore implies fullness of $\eta(\iota e_{11}) = |w|$. $\square$

5 From a cone map with a full corner to a dimension-drop map

Recall the unitized matrix cone

$$C = \{f \in C([0, 1], M_2) : f(0) \in \mathbb{C}1\}, \quad \iota(t) = t.$$

We show that a unital map from C whose first matrix corner is full produces a unital dimension-drop map after taking a finite tensor power. The comparison we construct has a fixed cutoff: the defect of an exactly equivalent orthogonal pair is Cuntz below the positive part of its first element minus $1/2$. Two elementary constructions in tensor powers of C will be useful. The first identifies these tensor powers with matrix-valued functions on a cube; the second moves a copy of M_2 continuously between the available tensor factors.

Lemma 5.1. *For $k \geq 1$, the algebra $C^{\otimes k}$ identifies with*

$$\left\{ f \in C([0, 1]^k, M_2^{\otimes k}) : f(t) \in \bigotimes_{i=1}^k E_i(t) \right\}, \quad E_i(t) = \begin{cases} M_2, & t_i > 0, \\ \mathbb{C}1, & t_i = 0. \end{cases}$$

Proof. The algebra C is nuclear, being the unitization of $C_0((0, 1]) \otimes M_2$. Its maximal and minimal tensor powers therefore agree. The natural map from its minimal tensor power into $C([0, 1]^k, M_2^{\otimes k})$ is isometric: the direct sum of point evaluations is a faithful representation of C , and its spatial tensor power is the direct sum of the evaluations on the cube. The range satisfies the indicated boundary conditions.

To prove the reverse inclusion, choose a vector-space basis b_0, b_1, b_2, b_3 of M_2 with $b_0 = 1$, and write a section as

$$f(t) = \sum_{\alpha \in \{0, 1, 2, 3\}^k} f_\alpha(t) b_{\alpha_1} \otimes \cdots \otimes b_{\alpha_k}.$$

Its scalar coefficients are continuous. If $S_\alpha = \{i : \alpha_i \neq 0\}$, the boundary conditions say that $f_\alpha(t) = 0$ whenever $t_i = 0$ for some $i \in S_\alpha$. Put $p_\alpha(t) = \prod_{i \in S_\alpha} t_i$, with empty product equal to one. Then

$$f_\alpha(t) \frac{p_\alpha(t)}{p_\alpha(t) + \varepsilon} \longrightarrow f_\alpha(t) \quad \text{uniformly as } \varepsilon \downarrow 0.$$

Indeed, f_α vanishes on the zero set of p_α , and is uniformly small on sufficiently small sublevel sets of p_α ; on their complements the displayed convergence is uniform. Thus f_α belongs to the closed scalar ideal generated by p_α .

Every continuous scalar function on the cube, multiplied by the matrix identity, lies in $C^{\otimes k}$, by approximation by finite sums of products of one-variable functions. Moreover,

$$p_\alpha(t) b_{\alpha_1} \otimes \cdots \otimes b_{\alpha_k}$$

is an elementary tensor of elements of C : its i -th factor is b_{α_i} if $i \in S_\alpha$, and is 1 otherwise. The scalar-ideal approximation now puts each summand of f , and hence f itself, in $C^{\otimes k}$. $\square$

Lemma 5.2. *For every $k \geq 1$, there is a point-norm continuous family of unital $*$ -homomorphisms*

$$\theta_r : M_2 \longrightarrow M_2^{\otimes k}, \quad r \in \Delta_{k-1} = \left\{ r \in [0, 1]^k : \sum_i r_i = 1 \right\},$$

whose range uses only the tensor factors indexed by $\{i : r_i > 0\}$. At the j -th vertex, θ_r is the j -th coordinate inclusion.

Proof. Let $F \in M_2 \otimes M_2$ be the flip unitary, and fix the unitary path

$$U(s) = \frac{1+F}{2} + e^{i\pi s} \frac{1-F}{2}, \quad 0 \leq s \leq 1.$$

Thus $U(0) = 1$, $U(1) = F$, and $F(a \otimes 1)F^* = 1 \otimes a$.

Proceed by induction on k , starting with the identity map when $k = 1$. For $r_k < 1$, let $\tau : M_2 \rightarrow M_2^{\otimes(k-1)}$ be the map supplied by induction for

$$\left(\frac{r_1}{1-r_k}, \dots, \frac{r_{k-1}}{1-r_k} \right),$$

and set

$$\theta_r(a) = (\tau \otimes \text{id})(U(r_k)(a \otimes 1)U(r_k)^*).$$

This is a unital $*$ -homomorphism, varying continuously where $r_k < 1$. At $r_k = 1$, define $\theta_r(a) = 1^{\otimes(k-1)} \otimes a$. For every preceding map τ , contractivity gives

$$\begin{aligned} \|\theta_r(a) - 1^{\otimes(k-1)} \otimes a\| &\leq \|U(r_k)(a \otimes 1)U(r_k)^* - F(a \otimes 1)F^*\| \\ &\leq 2\|a\| \|U(r_k) - F\|. \end{aligned}$$

The last quantity tends to zero as $r_k \rightarrow 1$, uniformly in τ . This proves continuity even when the normalized preceding weights have no limit.

If $r_k = 0$, the formula is exactly $\tau(a) \otimes 1$. If $r_i = 0$ for an earlier index, the induction hypothesis makes the i -th tensor factor scalar throughout the range of τ , and applying $\tau \otimes \text{id}$ preserves this restriction. These observations prove the support condition on every face, as well as the assertion at the vertices. $\square$

Proposition 5.3. *Let P be a nonzero unital C^* -algebra. Suppose that there is a unital $*$ -homomorphism $\eta : C \rightarrow P$ such that $\eta(\iota e_{11})$ is full in P . Then, for some finite $m \geq 1$, the maximal tensor power $B = P^{\otimes m}$ contains orthogonal positive contractions b_1, b_2 and an element v_B such that*

$$b_1 = v_B^* v_B, \quad b_2 = v_B v_B^*, \quad 1_B - b_1 - b_2 \precsim (b_1 - 1/2)_+.$$

Consequently there is a unital $$ -homomorphism*

$$I(2, 3) \longrightarrow P^{\otimes m}.$$

Proof. We first obtain a finite comparison constant from fullness. An auxiliary tensor power will provide an element whose class is both small enough to fit that many copies under the unit and large enough to dominate the unit after another finite amplification. The cube construction then places the defect under elements on which the first member of the pair acts as the identity.

A full cut and its comparison. Write $b = \eta(\iota e_{11})$. By fullness, choose $h_i, \ell_i \in P$, $1 \leq i \leq N$, such that

$$\left\| 1_P - \sum_{i=1}^N h_i b \ell_i \right\| < \frac{1}{2}.$$

For sufficiently small $0 < \delta < 1$, the element

$$c = (b - \delta)_+ = \eta(q), \quad q(t) = (t - \delta)_+ e_{11},$$

still satisfies $\|1_P - S\| < 1$, where $S = \sum_i h_i c \ell_i$. Thus S is invertible, and c is full.

We record an exact matrix comparison. Put

$$T = (h_1 c^{1/2}, \dots, h_N c^{1/2}), \quad Z = (c^{1/2} \ell_1, \dots, c^{1/2} \ell_N)^T, \quad d = Z^* Z.$$

Since $S = TZ$, we have

$$S^* S \leq \|T\|^2 d.$$

Invertibility of S therefore implies invertibility of d . The column $V = (\ell_i d^{-1/2})_{i=1}^N$ satisfies

$$V^* \text{diag}(c, \dots, c) V = 1_P, \quad \text{so} \quad [1_P] \leq N[c]. \quad (5.1)$$

An auxiliary amplification. Let $c_1 = c$ and $c_2 = \eta((\iota - \delta)_+ e_{22})$. These are orthogonal equivalent positive contractions, with equivalence witness $\eta(\sqrt{(\iota - \delta)_+} e_{21})$, and $c_1 + c_2 \leq 1_P$. Choose $L \geq 1$ such that $2^L \geq N$, and set

$$R = P^{\otimes L}, \quad e = c^{\otimes L}, \quad k = N^L.$$

The 2^L elements

$$c_{\sigma_1} \otimes \dots \otimes c_{\sigma_L}, \quad \sigma \in \{1, 2\}^L,$$

are pairwise orthogonal, all have Cuntz class $[e]$, and have sum $(c_1 + c_2)^{\otimes L} \leq 1_R$. Tensoring (5.1) L times gives the other bound in

$$N[e] \leq 2^L[e] \leq [1_R] \leq N^L[e] = k[e]. \quad (5.2)$$

The two integers play different roles: N is the comparison constant from fullness, whereas k will be the number of cube coordinates. Both are finite and may depend on P and η . After applying

$\eta^{\otimes k}$, the cube construction below will supply positive elements $X, Y \in P^{\otimes k}$ with $k[X] \leq N[Y]$. The two bounds on $[e]$ will then combine with this inequality to give $[X \otimes 1_R] \leq [Y \otimes 1_R]$.

An orthogonal pair in the cube. We next work in the cube model of $C^{\otimes k}$ from Lemma 5.1. Regard scalar functions as scalar matrix functions, and define

$$g(s) = (1 - 2s/\delta)_+, \quad H(t) = 1 - \prod_{j=1}^k g(t_j).$$

For $t \neq 0$, put

$$d(t) = \max_j t_j, \quad r_i(t) = \frac{(t_i/d(t) - 1/2)_+}{\sum_{j=1}^k (t_j/d(t) - 1/2)_+}.$$

At least one term in the denominator is $1/2$, so $r(t)$ is well defined and continuous off the origin. Let θ_r be the family from Lemma 5.2, and set

$$a_i(t) = H(t)\theta_{r(t)}(e_{ii}) \quad (i = 1, 2), \quad v(t) = \sqrt{H(t)}\theta_{r(t)}(e_{21}) \quad (t \neq 0).$$

Set all three functions equal to zero at the origin. The bounds

$$0 \leq H(t) \leq \frac{2}{\delta} \sum_j t_j, \quad \|a_i(t)\| \leq H(t), \quad \|v(t)\| \leq \sqrt{H(t)}$$

prove continuity there. If $t \neq 0$ and $t_j = 0$, then $r_j(t) = 0$, so the support condition on θ supplies the scalar boundary condition in that tensor factor. Consequently $a_1, a_2, v \in C^{\otimes k}$. They satisfy

$$v^*v = a_1, \quad vv^* = a_2, \quad a_1a_2 = 0, \quad a_1 + a_2 = H1. \quad (5.3)$$

In particular a_1, a_2 are equivalent orthogonal positive contractions, and their defect is

$$X = 1 - a_1 - a_2 = g^{\otimes k}.$$

For $1 \leq j \leq k$, define

$$y_j = g^{\otimes(j-1)} \otimes q \otimes g^{\otimes(k-j)}.$$

The y_j are positive and pairwise orthogonal, because $qg = 0$. Moreover,

$$a_1y_j = y_j \quad (1 \leq j \leq k). \quad (5.4)$$

Indeed, wherever $y_j(t) \neq 0$, one has $t_j > \delta$ and $t_i < \delta/2$ for every $i \neq j$. Hence $d(t) = t_j$ and $t_i/d(t) < 1/2$ for all $i \neq j$. The vector $r(t)$ is therefore exactly the j -th vertex, while $H(t) = 1$. At that vertex $\theta_{r(t)}(e_{11})$ acts as the identity on the q -factor of $y_j(t)$. At all other points with $y_j(t) = 0$, the desired identity is immediate. Figure 1 illustrates this localization in two coordinates.

Use the same letters for the images of these elements under $\eta^{\otimes k} : C^{\otimes k} \rightarrow P^{\otimes k}$, and put $Y = \sum_{j=1}^k y_j$. All the identities just proved persist. Writing $G = \eta(g)$, we have $[G] \leq [1_P] \leq N[c]$. Tensor this comparison with G in the other $k - 1$ positions to obtain

$$[X] \leq N[y_j] \quad (1 \leq j \leq k).$$

Adding these comparisons and using orthogonality gives

$$k[X] \leq N \sum_{j=1}^k [y_j] = N[Y]. \quad (5.5)$$

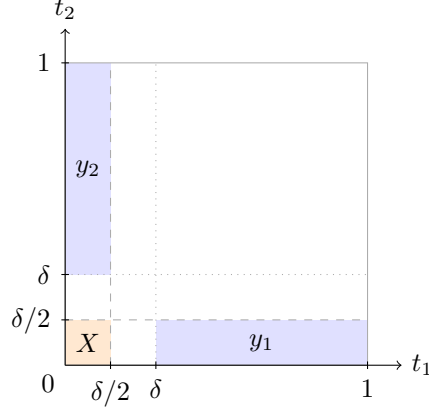


Figure 1: Support geometry for $k = 2$, drawn schematically. Shading shows the closures of the regions where X , y_1 , and y_2 can be nonzero. In the y_j region, $r(t)$ is the j -th vertex and $H(t) = 1$, giving (5.4).

Thus the cube supplies k copies of the defect below N copies of an element supported where $a_1 = 1$. The auxiliary algebra R was chosen precisely to remove these unequal multiplicities without any cancellation in the Cuntz order.

Absorbing the defect. Finally, work in

$$B = P^{\otimes k} \otimes R = P^{\otimes(k+L)}, \quad b_i = a_i \otimes 1_R.$$

Since $a_1 Y = Y$, taking adjoints gives $Y a_1 = Y$. Thus a_1 and Y commute, and functional calculus gives $(a_1 - 1/2)_+ Y = Y/2$. In particular the element $z = \sqrt{2Y} \otimes 1_R$ satisfies

$$z^*(b_1 - 1/2)_+ z = Y \otimes 1_R.$$

The following use of tensor divisibility to obtain comparison follows the pattern of Robert and Rørdam [RR, Lemma 6.1(i)]; we verify the comparisons directly in maximal tensor products. Together with (5.2) and (5.5), this yields

$$\begin{aligned} [X \otimes 1_R] &\leq k[X \otimes e] \\ &\leq N[Y \otimes e] \\ &\leq [Y \otimes 1_R] \\ &\leq [(b_1 - 1/2)_+]. \end{aligned}$$

The first comparison tensors $[1_R] \leq k[e]$ with X , the second tensors (5.5) with e , and the third tensors $N[e] \leq [1_R]$ with Y . The last uses the exact witness z . These are comparisons in finite matrices over the maximal tensor product B .

To express the conclusion inside B itself, recall that if $a, b \in B_+$ and

$$W_n^*(b \oplus 0)W_n \longrightarrow a \oplus 0$$

in finite matrices, compression to the upper-left corner gives $(W_n)_{11}^* b (W_n)_{11} \rightarrow a$. The final comparison therefore gives

$$1_B - b_1 - b_2 = X \otimes 1_R \precsim (b_1 - 1/2)_+ \quad \text{in } B.$$

By (5.3), the elements b_1, b_2 are orthogonal positive contractions with exact equivalence witness $v_B = v \otimes 1_R \in B$. This proves the asserted pair and defect comparison. Proposition 2.1, with cutoff $1/2$, now gives a unital $*$ -homomorphism $I(2, 3) \rightarrow B$. Taking $m = k + L$ completes the proof. $\square$

6 From finite tensor powers to absorption

Proof of Theorem 1.2. Proposition 4.7 supplies a finite tensor power $P = D^{\otimes m_0}$ and a unital map $\eta : C \rightarrow P$ for which $\eta(\iota e_{11})$ is full. Proposition 5.3 supplies a unital map from $I(2, 3)$ into a finite tensor power $P^{\otimes s}$. Associativity of the maximal tensor product identifies the latter with $D^{\otimes m_0 s}$, as required. $\square$

For the application, we first record explicitly why the possibly nonseparable central-sequence algebra contains a suitable separable subalgebra. This is the separable-reduction argument of [KR, Lemma 3.5]; we include its finite commutator witness.

Lemma 6.1. *Every nonzero unital C^* -algebra B without characters contains a separable C^* -subalgebra D without characters and with $1_D = 1_B$.*

Proof. The closed ideal generated by commutators of selfadjoint elements is B ; a proper such ideal would give a nonzero unital commutative quotient and hence a character. Consequently, there are finitely many $h_i, k_i \in B$ and $b_i, c_i \in B_{\text{sa}}$ such that

$$\left\| 1_B - \sum_{i=1}^r h_i [b_i, c_i] k_i \right\| < 1.$$

Let D be generated by 1_B and all the displayed elements. It is unital and separable. A character of D would send the sum to zero and the unit to one, contradicting its contractivity and the strict norm bound. $\square$

Proof of Theorem 1.1. Assume that $F_\omega(A)$ has no characters. This algebra is nonzero and unital because it contains the diagonal unit of A . By Lemma 6.1, choose a separable unital characterless subalgebra $D \subset F_\omega(A)$ with the same unit. By Theorem 1.2 and Proposition 2.2(1), there is a composition of unital $*$ -homomorphisms

$$I(2, 3) \longrightarrow D^{\otimes \max m} \longrightarrow F_\omega(A).$$

Denote its image by E . The image is separable and is a quotient of $I(2, 3)$. Every irreducible representation of $I(2, 3)$ factors through evaluation at a point of $[0, 1]$: the central algebra $C([0, 1])$ acts by a character, and sections vanishing at its evaluation point are annihilated, by approximation using scalar functions vanishing there. The resulting irreducible dimensions are 2 at 0, 3 at 1, and 6 in the interior. Thus $I(2, 3)$ is subhomogeneous and has no characters. The same properties hold for E , since any irreducible representation or character of E pulls back along the quotient map. Also $1_E = 1_{F_\omega(A)}$. Proposition 2.2(2) gives $A \cong A \otimes_{\min} \mathcal{Z}$.

Conversely, if $A \cong A \otimes_{\min} \mathcal{Z}$, Proposition 2.2(2) supplies a unital subalgebra $E \subset F_\omega(A)$ without characters. Any character of $F_\omega(A)$ would restrict to a character of E , a contradiction. $\square$

Proof of Corollary 1.3. Because D is nonzero and unital, each spatial connecting map $x \mapsto x \otimes 1_D$ is a unital embedding. Thus B is nonzero, unital, and separable, and the canonical maps

$\iota_m : D^{\otimes_{\min} m} \hookrightarrow B$ are unital. Choose the unital homomorphism ϕ from Theorem 1.2 and compose it with the canonical maximal-to-minimal quotient and the finite-stage inclusion:

$$I(2, 3) \xrightarrow{\phi} D^{\otimes_{\max} m} \xrightarrow{q_m} D^{\otimes_{\min} m} \xrightarrow{\iota_m} B.$$

The quotient q_m is unital, with no nuclearity assumption on D . Let θ denote this composite and put $E = \theta(I(2, 3))$. Since θ is unital, $1_E = 1_B \neq 0$. The algebra $I(2, 3)$ is separable and, as checked above, subhomogeneous and characterless. Its quotient $E \cong I(2, 3)/\ker \theta$ is therefore separable and subhomogeneous: every irreducible representation of E pulls back to one of $I(2, 3)$. A character of E would compose with the unital quotient map to a character of $I(2, 3)$, which is impossible. These conclusions hold even when θ has a kernel.

Dadarlat–Toms [DT, Theorem 1.1], applied to the unital separable algebra D and this same-unit subalgebra of B , gives $B \cong B \otimes_{\min} \mathcal{Z}$. Finally, because $B \neq 0$, the map $z \mapsto 1_B \otimes z$ embeds $\mathcal{Z}$ unitaly into $B \otimes_{\min} \mathcal{Z}$. Transporting this map through the inverse of an isomorphism $B \cong B \otimes_{\min} \mathcal{Z}$, which is automatically unital, gives the asserted unital copy of $\mathcal{Z}$ in B . $\square$

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